CAPM and APT Models

Equilibrium and Arbitrage Approaches to Expected Return

IEOR E4576 — Data-Driven Models in Finance — Fall 2026 — Week 3

Contents

1 Capital Asset Pricing Model (CAPM) 1
1.1 CAPM Assumptions . . . 1
1.2 CAPM Model . . . 1

2 Arbitrage Pricing Theory (APT) 2
2.1 APT Assumptions . . . 2
2.2 APT Model . . . 2

3 Key Differences Between CAPM and APT 3

1 Capital Asset Pricing Model (CAPM)

The Capital Asset Pricing Model (CAPM) is a fundamental model in finance that describes the relationship between systematic risk and expected return for assets, particularly stocks.

1.1 CAPM Assumptions

1. **Homogeneous expectations.** All investors have identical expectations about expected returns, variances, and covariances of all assets.
2. **Mean-variance optimisation.** All investors are mean-variance optimisers who choose portfolios based solely on expected return and variance.
3. **Single-period model.** All investors have the same single-period investment horizon.
4. **Risk-free asset.** There exists a risk-free asset with known return r_f, and investors can borrow and lend unlimited amounts at this rate.
5. **Perfect capital markets.** No transaction costs; no taxes; perfect divisibility of assets; perfect liquidity; no restrictions on short selling.
6. **Price-taking behaviour.** All investors are price takers; no single investor can influence market prices.
7. **Unlimited borrowing and lending.** Investors can borrow and lend unlimited amounts at the risk-free rate.

1.2 CAPM Model

The CAPM is a statement about **excess returns** — returns in excess of the risk-free rate, $R_i^e = R_i - r_f$. Under the CAPM assumptions, the expected excess return of any asset i is given by

$$E[R_i] - r_f = \beta_i \left(E[R_m] - r_f\right) \tag{1}$$

where

- $E[R_i] - r_f$ — expected excess return of asset i;
- r_f — risk-free rate of return;
- β_i — beta of asset i, measuring systematic risk;
- $E[R_m]$ — expected return of the market portfolio. Under the CAPM assumptions the market portfolio is the capitalisation-weighted portfolio of all investable assets; in practice it is typically proxied by a cap-weighted index such as the S&P 500;
- $(E[R_m] - r_f)$ — market risk premium.

The beta coefficient is defined as

$$\beta_i = \frac{\mathrm{Cov}(R_i, R_m)}{\mathrm{Var}(R_m)} = \frac{\sigma_{i,m}}{\sigma_m^2}, \tag{2}$$

where $\sigma_{i,m}$ is the covariance between asset i and the market portfolio, and σ_m^2 is the variance of the market portfolio.

Excess returns in practice. With a constant r_f the excess-return form is just a rearrangement of $E[R_i] = r_f + \beta_i(E[R_m] - r_f)$. In data r_f varies period to period, so the CAPM is estimated as a time-series regression of excess returns on the market excess return,

$$R_{i,t} - r_{f,t} = \alpha_i + \beta_i \left(R_{m,t} - r_{f,t}\right) + \epsilon_{i,t}, \tag{3}$$

and β_i is the covariance of the excess returns divided by the variance of the market excess return. The CAPM's testable prediction is $\alpha_i = 0$ for every asset: the market excess return alone explains expected excess returns.

Without a risk-free asset. Black's (1972) zero-beta CAPM drops Assumption 4 and replaces r_f with $E[R_z]$, the expected return on a portfolio uncorrelated with the market: $E[R_i] - E[R_z] = \beta_i \left(E[R_m] - E[R_z]\right)$.

2 Arbitrage Pricing Theory (APT)

The Arbitrage Pricing Theory (APT), developed by Stephen Ross, provides a multi-factor approach to asset pricing that generalises the single-factor CAPM.

2.1 APT Assumptions

The APT separates a *statistical* assumption about how returns move from an *economic* assumption (no arbitrage), and derives a restriction on expected returns from the two.

1. **Factor structure of returns.** Realised returns are driven by k common shocks plus asset-specific noise:

$$R_i = a_i + \sum_{j=1}^{k} \beta_{ij} F_j + \epsilon_i, \qquad i = 1, \ldots, N, \tag{4}$$

   where the factors F_j have mean zero, so $a_i = E[R_i]$, and $k \ll N$. The intercepts a_i are *unrestricted*: this assumption describes the covariance structure of returns and says nothing about what expected returns should be.

2. **Diversifiable idiosyncratic risk.** $E[\epsilon_i] = 0$, $\mathrm{Cov}(\epsilon_i, F_j) = 0$ for all i, j, and the ϵ_i are at most weakly correlated across assets (uncorrelated is the simplest case), so idiosyncratic risk vanishes in large, well-diversified portfolios.

3. **No arbitrage.** There is no portfolio that costs nothing, carries no risk, and earns a positive payoff.

4. **Frictionless trading.** Investors can trade any combination of assets, including short positions, without transaction costs.

5. **Risk-free asset.** There exists a risk-free asset with return r_f. As with the CAPM, this is a convenience rather than a necessity (see below). Unlike the CAPM, nothing is assumed about investor preferences, common expectations, or the market portfolio.

2.2 APT Model

The arbitrage argument. Because there are many more assets than factors, one can form a zero-cost portfolio (weights summing to zero) whose factor exposures all cancel; with many assets its idiosyncratic risk diversifies away (Assumption 2). Such a portfolio is riskless and costless, so by Assumption 3 its expected payoff must be zero. This holds for every such portfolio only if the intercepts a_i in (4) lie on a common line in the loadings. A well-diversified unit-cost portfolio with zero exposure to every factor is riskless, so it must earn r_f; this pins the intercept of the line at r_f.

The APT pricing restriction. The expected excess return of asset i is

$$E[R_i] - r_f = \sum_{j=1}^{k} \beta_{ij} \lambda_j \tag{5}$$

where

- $E[R_i] - r_f$ — expected excess return of asset i;

- r_f — risk-free rate of return;
- β_{ij} — factor loading (sensitivity) of asset i to factor j, from (4);
- λ_j — risk premium for factor j, the same for every asset: the expected excess return of a portfolio with unit exposure to factor j and zero exposure to the others;
- k — number of systematic factors.

Assumption 1 lets each a_i be anything. The APT's content is the restriction $a_i = r_f + \sum_j \beta_{ij}\lambda_j$: assets with the same factor exposures must have the same expected return.

An approximate result. With idiosyncratic risk present, the arbitrage argument works only in the limit of many assets, so the restriction holds for *most* assets as $N \to \infty$ rather than exactly for each one.

Excess returns in practice. When the factors are excess returns on traded portfolios (e.g. the Fama–French factors), the APT is estimated as a time-series regression of excess returns,

$$R_{i,t} - r_{f,t} = \alpha_i + \sum_{j=1}^{k} \beta_{ij} f_{j,t} + \epsilon_{i,t}, \tag{6}$$

where $f_{j,t}$ is the factor's excess return, and the APT predicts $\alpha_i = 0$.

Without a risk-free asset. Ross's (1976) original statement drops Assumption 5: $E[R_i] = \lambda_0 + \sum_j \beta_{ij}\lambda_j$, where λ_0 is the expected return on a portfolio with zero exposure to every factor (the zero-beta rate). With a risk-free asset, $\lambda_0 = r_f$ and this is (5).

Relation to the CAPM. With $k = 1$ and the market as the single factor, $\lambda_1 = E[R_m] - r_f$ and the APT restriction has the same form as the CAPM. The difference is in what is derived: the CAPM *concludes* from investor equilibrium that the market is the priced factor, while the APT takes the factors as given and does not say what they are.

3 Key Differences Between CAPM and APT

Aspect	CAPM	APT
Risk factors	Single (market)	Multiple factors
Assumptions	More restrictive	More flexible
Testability	More testable	Less testable
Practical implementation	Simpler	More complex
Market portfolio	Required	Not required

Both models provide frameworks for understanding the relationship between risk and expected return, with the APT offering greater flexibility through its multi-factor approach while the CAPM provides clearer theoretical foundations and easier implementation.